

METHODOLOGY AND EXPERIMENTS

A. Aim of Study

The primary aim of this study is to develop a machine learning framework capable of accurately predicting whether a reported crime incident in the City of Chicago will result in an arrest. Unlike many existing studies that primarily focus on crime occurrence prediction or hotspot detection, this research emphasizes arrest prediction by integrating multiple heterogeneous data sources, including historical crime records, socio-economic indicators, weather conditions, and public transportation mobility data. By combining these diverse data sources, the proposed framework seeks to provide a more comprehensive understanding of the factors that influence arrest outcomes and improve the predictive capability of machine learning models.

The study investigates how spatial, temporal, environmental, and socio-economic characteristics influence the likelihood of an arrest following a reported crime incident. To achieve this objective, multiple machine learning and deep learning algorithms are developed, trained, and evaluated using balanced datasets to determine the most effective approach for arrest prediction.

The objectives of this study are to develop a comprehensive framework for predicting arrest outcomes by first preprocessing, cleaning, and integrating multiple heterogeneous datasets into a unified dataset suitable for machine learning analysis. The study then aims to investigate the influence of crime-related, spatial, temporal, socio-economic, weather, and public mobility factors on arrest prediction by identifying the most informative features contributing to the classification task. Given the inherent class imbalance in the arrest data, appropriate sampling techniques are employed to improve the learning capability of the predictive models. Subsequently, a range of machine learning and deep learning algorithms are developed and trained using a consistent experimental framework to enable a fair comparison of their predictive performance. The trained models are evaluated using widely accepted classification metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC, to assess their effectiveness in predicting arrest outcomes. Finally, the study seeks to identify the most reliable and robust predictive model capable of supporting data-driven decision-making in crime analysis and law enforcement.

B. Response (Dependent) and Independent Variables

The predictive modeling framework developed in this study is based on a binary classification problem in which the objective is to predict whether a reported crime incident results in an arrest. Accordingly, the dataset consists of one response (dependent) variable and multiple independent (predictor) variables obtained by integrating crime records with socio-economic, weather, public transportation mobility, spatial, and temporal information. These variables were selected to capture

different aspects of crime incidents that may influence arrest outcomes.

Response (Dependent) Variable

The response variable represents the target outcome that the machine learning and deep learning models aim to predict. In this study, the Arrest attribute from the Chicago crime dataset is used as the dependent variable. It is a binary variable indicating whether an arrest was made following a reported crime incident.

Variable	Description	Values
Arrest	Indicates whether an arrest was made for the reported crime incident.	0 = No Arrest, 1 = Arrest

Table 1: Response (Dependent) Variable

Independent Variables

The independent variables represent the predictor variables used to train the classification models. These variables were obtained from multiple data sources and are grouped into six categories based on their characteristics.

Crime-Related Variables

These variables describe the characteristics of the reported crime incident.

Variable	Description
Primary Type	Primary category of the reported crime.
Description	Detailed description of the crime.
Location Description	Type of location where the crime occurred.
Domestic	Indicates whether the crime was domestic-related.
FBI Code	FBI crime classification code.
Beat	Police beat responsible for the incident.
District	Police district where the crime occurred.
Community Area	Community area in which the crime occurred.

Table 2: Crime-Related Variables

Spatial Variables

These variables represent the geographical location of each crime incident.

Variable	Description
Latitude	Geographic latitude of the crime location.
Longitude	Geographic longitude of the crime location.

Table 3: Spatial Variables

Temporal Variables

These variables capture the temporal characteristics of crime incidents.

Variable	Description
Year	Year in which the crime occurred.
Month	Month of occurrence.
Day	Day of the month.
Day of Week	Day of the week on which the crime occurred.
Hour	Hour at which the crime was reported.

Table 4: Temporal Variables

Socio-economic Variables

These variables describe the socio-economic conditions of the community where the crime occurred.

Variable	Description
Per Capita Income	Average income per person within the community area.

Unemployment Rate	Percentage of unemployed individuals in the community area.
No High School Percentage	Percentage of residents without a high school diploma.

Table 5: Socio-economic Variables

Weather Variables

Weather conditions corresponding to the date of each crime incident were incorporated into the dataset.

Variable	Description
TMAX (Maximum Temperature)	Maximum daily temperature (°C).
TMIN (Minimum Temperature)	Minimum daily temperature (°C).
PRCP (Rainy)	Daily precipitation.
AWND (Average Wind Speed)	Average daily wind speed.
Average Temperature	Mean daily temperature calculated from TMAX and TMIN.

Table 6: Weather Variables

Public Transportation Mobility Variables

Daily public transportation ridership data were included to represent human mobility within the city.

Variable	Description
Bus Ridership	Total daily CTA bus boardings.

Rail Ridership	Total daily CTA rail boardings.
Total Ridership	Combined daily bus and rail ridership.

Table 7: Public Transportation Mobility Variables

The final predictive dataset used for model development consists of one dependent variable (Arrest) and twenty-seven independent variables representing crime characteristics, spatial location, temporal information, socio-economic conditions, weather conditions, and public transportation mobility. The integration of these heterogeneous variables enables the framework to capture multiple dimensions influencing arrest outcomes and provides a comprehensive basis for predictive modeling.

C. Factors and Levels

In experimental research, factors are the independent components that are intentionally varied to investigate their effect on the response variable, while levels represent the different settings or values associated with each factor. In this study, several experimental factors were considered to evaluate their influence on the performance of arrest prediction models. These factors include the data balancing strategy, learning algorithm, feature set, and hyperparameter configuration. The corresponding levels for each factor are summarized in Table 8.

Factor	Levels	Purpose
Data Balancing Strategy	Original (Imbalanced), Random Under-Sampling, Random Over-Sampling	To investigate the impact of class balancing techniques on arrest prediction performance.
Learning Algorithm	Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, Linear SVM, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Temporal Convolutional Network (TCN)	To compare the predictive performance of traditional machine learning and deep learning algorithms.

Feature Set	Crime-related Features, Spatial Features, Temporal Features, Socio-economic Features, Weather Features, Public Transportation Mobility Features	To incorporate multiple dimensions that may influence arrest outcomes.
Hyperparameter Configuration	Algorithm-specific optimized hyperparameters	To improve model performance while reducing overfitting and enhancing generalization.

Table 8: Experimental Factors and Levels

Among these factors, the data balancing strategy plays a critical role because the original dataset exhibits a significant class imbalance. In the final dataset, 74.20% of the crime incidents resulted in no arrest, while only 25.80% resulted in an arrest. Such an imbalanced class distribution can bias machine learning algorithms toward the majority class, leading to poor identification of arrest cases. To mitigate this issue, Random Under-Sampling and Random Over-Sampling techniques were applied, and their effectiveness was systematically compared with the original imbalanced dataset to evaluate their impact on the predictive performance of the classification models.

The learning algorithm constitutes another important experimental factor. A diverse set of traditional machine learning algorithms and deep learning models was evaluated using the same training and testing datasets to ensure a fair comparison. Each algorithm was trained using its optimized hyperparameter configuration to maximize predictive performance.

The feature set represents the collection of predictor variables used for model development. Instead of relying solely on crime-related information, this study integrates spatial, temporal, socio-economic, weather, and public transportation mobility variables to capture the multifaceted nature of crime incidents and their potential influence on arrest outcomes.

Finally, algorithm-specific hyperparameters were optimized experimentally to enhance classification accuracy and improve the generalization capability of each model. The combined evaluation of these experimental factors provides a comprehensive assessment of the proposed arrest prediction framework and facilitates the identification of the most effective predictive model.

D. Experimental Design

The experimental design adopted in this study consists of a series of sequential stages that

systematically transform raw crime data into a comprehensive machine learning framework for arrest prediction. Each stage was carefully designed to ensure data quality, reproducibility, and consistency throughout the experimental process. Furthermore, the same experimental procedure was applied across all machine learning and deep learning models to ensure that performance comparisons were fair and unbiased.

a) Data Collection and Integration

The first stage of the experimental design involved collecting data from multiple publicly available sources. The primary dataset consisted of historical crime records obtained from the Chicago Crime Dataset, which contains detailed information regarding crime incidents, including crime type, location, arrest status, police jurisdiction, and incident time. Since arrest outcomes are influenced by multiple external factors, three additional datasets were incorporated into the study: socio-economic indicators describing community characteristics, daily weather observations obtained from NOAA, and daily public transportation ridership data provided by the Chicago Transit Authority (CTA).

Following data collection, the datasets were integrated into a unified analytical dataset. Crime incidents were linked with weather observations and mobility data based on the incident date, while socio-economic variables were associated using the corresponding community area. This integration enabled each crime record to contain not only crime-specific information but also environmental, demographic, and mobility-related characteristics, thereby providing a richer representation of the conditions under which crimes occurred.

b) Data Preprocessing

The integrated dataset underwent extensive preprocessing to improve data quality and ensure consistency before model development. Duplicate records were removed, missing values were identified and handled appropriately, and inconsistencies across the merged datasets were corrected. Missing geographical information, including community areas and police districts, was reconstructed using spatial relationships and existing administrative information to maximize data completeness.

Several temporal features were also derived from the original crime date, including year, month, day, day of the week, and hour, to capture temporal crime patterns. Additional weather-related variables, such as average temperature, were computed from the original weather observations. Since several predictor variables were categorical in nature, including crime type, location description, and FBI crime code, Label Encoding was applied to convert these variables into numerical values suitable for machine learning algorithms while preserving their categorical distinctions.

c) Feature Selection

Feature selection was performed to identify the predictor variables that contributed most significantly to arrest prediction while reducing unnecessary complexity in the dataset. Initially, a comprehensive set of variables representing crime characteristics, spatial information, temporal patterns, socio-economic conditions, weather observations, and public transportation mobility was considered.

A Random Forest-based feature importance analysis was conducted to estimate the relative contribution of each variable to the prediction task. Variables with higher importance scores were retained, whereas less informative variables were excluded from the final modeling process. This approach reduced data dimensionality, improved computational efficiency, minimized redundancy among predictor variables, and enhanced the interpretability of the resulting models. The final dataset consisted of twenty-seven independent variables together with the binary response variable representing arrest outcomes.

d) Dataset Partitioning

To ensure an unbiased evaluation of predictive performance, the final dataset was divided into separate training and testing subsets using an 80:20 ratio. The training dataset was used for model learning, whereas the testing dataset remained completely unseen during training and was reserved exclusively for performance evaluation.

A stratified sampling strategy was employed during dataset partitioning to preserve the original class distribution in both subsets. This approach ensured that the proportion of arrest and non-arrest cases remained consistent between the training and testing datasets, thereby producing a more reliable estimate of model generalization on unseen crime incidents.

e) Handling Class Imbalance

An important challenge encountered during model development was the highly imbalanced distribution of the response variable. Approximately 74.20% of the crime incidents resulted in no arrest, whereas only 25.80% resulted in an arrest. Such an imbalance can bias machine learning algorithms toward the majority class, causing poor predictive performance for the minority arrest class.

To investigate the impact of class balancing on model performance, three experimental scenarios were considered. The first scenario used the original imbalanced dataset without modification. The second scenario applied Random Under-Sampling to reduce the size of the majority class, while the third scenario employed Random Over-Sampling to increase the representation of the minority class. Comparing these three strategies enabled the study to evaluate the effectiveness of sampling techniques in improving the identification of arrest cases while maintaining overall classification performance.

f) Model Development and Training

Following data preparation, a comprehensive set of predictive models was developed to evaluate different learning strategies for arrest prediction. The machine learning models included Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, Linear Support Vector Machine (Linear SVM), and Multi-Layer Perceptron (MLP). In addition, three deep learning architectures—Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN)—were implemented to investigate their capability in learning complex relationships among predictor variables.

Each algorithm was trained using the same training dataset to ensure consistency throughout the experiments. Algorithm-specific hyperparameters were optimized through experimental tuning to improve predictive accuracy while reducing the risk of overfitting. Applying identical training and testing datasets for every model ensured that differences in performance were attributable to the learning algorithms themselves rather than variations in the experimental setup.

g) Model Evaluation

After model training, the predictive performance of each algorithm was evaluated using the independent testing dataset. Multiple classification metrics were employed to provide a comprehensive assessment of model effectiveness, including Accuracy, Precision, Recall, F1-score, Receiver Operating Characteristic Area Under the Curve (ROC-AUC), Precision-Recall Area Under the Curve (PR-AUC), and confusion matrices.

Because arrest prediction represents an imbalanced binary classification problem, particular emphasis was placed on Recall, F1-score, ROC-AUC, and PR-AUC rather than relying solely on overall accuracy. These evaluation metrics provide a more balanced assessment of each model's ability to correctly identify arrest cases while minimizing false predictions.

h) Model Comparison and Selection

The final stage of the experimental design involved comparing the predictive performance of all machine learning and deep learning models under identical experimental conditions. The comparison considered multiple evaluation metrics together with the models' robustness, generalization capability, and computational efficiency.

Based on the comparative analysis, the model demonstrating the best overall balance between predictive accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC was selected as the final arrest prediction model. This systematic evaluation ensured that the selected model was not only accurate but also reliable for predicting arrest outcomes across a diverse range of crime incidents.

E. Experiment Performance and Revisions

The development of the arrest prediction framework was carried out through a sequence of experiments, with each experiment addressing a specific objective in the model development process. The outcome of one experiment provided the foundation for subsequent revisions, enabling continuous improvements in the predictive capability of the classification models. This iterative experimental approach facilitated a systematic evaluation of data preparation techniques, feature engineering, feature selection, class balancing strategies, hyperparameter optimization, and algorithm performance. The experimental progression is described as follows.

a) Experiment 1: Baseline Model Using the Original Integrated Dataset

The first experiment established the baseline performance of the arrest prediction framework using the original integrated dataset after data preprocessing and integration. The dataset contained all meaningful predictor variables, including crime-related, spatial, temporal, socio-economic, weather, and public transportation mobility features, while preserving the original imbalanced class distribution. Several baseline machine learning models were trained using this dataset to establish benchmark performance for subsequent experiments.

Although the baseline models achieved satisfactory overall accuracy, detailed analysis revealed that they were biased toward the majority class due to the inherent imbalance between arrest and non-arrest incidents. Consequently, the models demonstrated comparatively lower recall and F1-score for the minority arrest class, indicating the need for further refinement of the predictive framework.

b) Experiment 2: Feature Importance Analysis using Random Forest

The second experiment focused on identifying the relative importance of each predictor variable in the arrest prediction task. A Random Forest classifier was employed to compute feature importance scores for all available predictor variables within the integrated dataset.

The resulting importance scores provided valuable insights into the contribution of crime-related, spatial, temporal, socio-economic, weather, and mobility variables toward arrest prediction. These findings established a quantitative basis for selecting the most informative variables in the subsequent experiment.

c) Experiment 3: Feature Selection

Based on the feature importance scores obtained in Experiment 2, the top 27 predictor variables with the highest importance scores were selected for subsequent model development. These variables were chosen because they contributed most significantly to arrest prediction while reducing redundancy and computational complexity.

d) Experiment 4: Baseline Model Using Selected Features

Following feature selection, a new baseline experiment was conducted using the selected 27 predictor variables while maintaining the original imbalanced class distribution. This experiment was designed to evaluate the impact of feature selection alone on classification performance before introducing any sampling techniques.

e) Experiment 5: Performance Improvement Using Random Under-Sampling

The fifth experiment investigated the effectiveness of Random Under-Sampling in addressing the class imbalance problem. The majority class was randomly reduced to create a more balanced training dataset while preserving the selected predictor variables.

This experiment evaluated whether reducing the dominance of the majority class could improve the models' ability to correctly identify arrest cases. The performance obtained from the under-sampled dataset was subsequently compared with the baseline models developed in Experiment 4.

f) Experiment 6: Performance Improvement Using Random Over-Sampling

The sixth experiment evaluated the effectiveness of Random Over-Sampling as an alternative strategy for handling class imbalance. Unlike under-sampling, this technique increased the representation of the minority arrest class while retaining all observations from the majority class.

The resulting balanced dataset enabled the classification models to learn more representative patterns associated with arrest incidents. The predictive performance achieved using Random Over-Sampling was compared with both the original imbalanced dataset and the Random Under-Sampling approach to determine the most suitable sampling strategy.

g) Experiment 7: Hyperparameter Optimization

After identifying the most effective sampling strategy, the hyperparameters of the selected machine learning and deep learning algorithms were optimized experimentally. Algorithm-specific parameters, including the number of estimators, learning rate, maximum tree depth, batch size, training epochs, optimizer settings, and other model-specific configurations, were adjusted to improve predictive performance and reduce the likelihood of overfitting.

The optimized configurations enhanced the stability, robustness, and generalization capability of the developed models while ensuring that each algorithm was evaluated under its most appropriate settings.

h) Experiment 8: Algorithm Comparison and Final Model Selection

The final experiment involved a comprehensive comparison of all optimized machine learning and deep learning models developed throughout the study. Every algorithm was evaluated using the same testing dataset and assessed using multiple classification performance metrics, including

Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC, and confusion matrices.

The comparative analysis considered not only predictive performance but also model robustness, computational efficiency, and generalization capability. Based on the overall evaluation, the most appropriate predictive model was selected for the proposed arrest prediction framework. The detailed results and justification for the final model selection are presented in the subsequent experimental results chapter.

F. Measuring Classification Performance

The performance of the developed machine learning and deep learning models was evaluated using several widely accepted classification metrics. Since arrest prediction is formulated as a binary classification problem with an imbalanced class distribution, relying solely on overall accuracy may provide a misleading assessment of model performance. Therefore, multiple evaluation metrics were employed to comprehensively assess the predictive capability of each model. These metrics evaluate different aspects of classification performance, including the ability to correctly identify arrest cases, minimize false predictions, and distinguish between the two classes.

Before defining the evaluation metrics, the outcomes of binary classification are represented using a confusion matrix. The confusion matrix summarizes the relationship between the actual and predicted class labels and forms the basis for calculating all classification performance metrics.

a) Confusion Matrix

A confusion matrix is a tabular representation of the prediction results obtained from a classification model. It consists of four possible outcomes:

Actual Class	Predicted: Arrest (1)	Predicted: No Arrest (0)
Arrest (1)	True Positive (TP)	False Negative (FN)
No Arrest (0)	False Positive (FP)	True Negative (TN)

Table 9: Confusion Matrix

where:

- **True Positive (TP):** The model correctly predicts that a crime incident results in an arrest.
- **True Negative (TN):** The model correctly predicts that a crime incident does not result in an arrest.

- **False Positive (FP):** The model incorrectly predicts an arrest when no arrest actually occurred.
- **False Negative (FN):** The model incorrectly predicts no arrest when an arrest actually occurred.

These four values serve as the foundation for calculating all subsequent evaluation metrics.

b) Accuracy

Accuracy measures the overall proportion of correctly classified observations among all crime incidents. It provides a general indication of model performance; however, for imbalanced datasets, accuracy alone may be misleading because a model can achieve high accuracy by predominantly predicting the majority class.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

c) Precision

Precision measures the proportion of correctly predicted arrest cases among all observations predicted as arrests. A high precision value indicates that when the model predicts an arrest, the prediction is likely to be correct.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Higher precision reduces the number of false positive predictions.

d) Recall (Sensitivity)

Recall, also referred to as sensitivity or the true positive rate, measures the proportion of actual arrest cases that are correctly identified by the model. This metric is particularly important in arrest prediction because failing to identify actual arrest cases may reduce the effectiveness of predictive policing strategies.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

A higher recall value indicates that the model successfully identifies a larger proportion of arrest cases.

e) F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of classification performance by considering both false positive and false negative predictions simultaneously. Because arrest prediction involves an imbalanced dataset, the F1-score is considered one of the most informative evaluation metrics in this study.

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (4)$$

A higher F1-score indicates a better balance between precision and recall.

f) Receiver Operating Characteristic Area Under the Curve (ROC-AUC)

The Receiver Operating Characteristic (ROC) curve illustrates the relationship between the True Positive Rate (Recall) and the False Positive Rate at different classification thresholds. The Area Under the ROC Curve (ROC-AUC) measures the model's ability to distinguish between arrest and non-arrest cases.

An ROC-AUC value of 1.0 represents perfect classification, whereas a value of 0.5 indicates performance equivalent to random guessing. Higher ROC-AUC values indicate better discrimination between the two classes.

g) Precision–Recall Area Under the Curve (PR-AUC)

The Precision–Recall (PR) curve illustrates the relationship between precision and recall across different decision thresholds. The Area Under the Precision–Recall Curve (PR-AUC) summarizes this relationship into a single performance measure.

PR-AUC is particularly useful for imbalanced classification problems because it focuses on the performance of the minority class. Since arrest cases represent only 25.80% of the observations in this study, PR-AUC provides a more informative evaluation of model performance than accuracy alone.

h) Importance of Multiple Evaluation Metrics

No single evaluation metric can completely describe the performance of a classification model, particularly when the dataset is imbalanced. Therefore, this study evaluates every machine learning and deep learning model using Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC to

obtain a comprehensive assessment of predictive performance.

Among these metrics, particular emphasis is placed on Recall, F1-score, ROC-AUC, and PR-AUC because they provide a more reliable evaluation of the model’s ability to correctly identify arrest cases while maintaining a balanced trade-off between false positive and false negative predictions. The combined use of these evaluation metrics ensures a fair and comprehensive comparison of all models developed in this research.

G. Algorithm Comparison and Selection

A comprehensive comparative analysis was conducted to evaluate the performance of multiple machine learning and deep learning algorithms for arrest prediction. The primary objective of this comparison was to identify the model that provides the most accurate, reliable, and robust prediction of arrest outcomes while maintaining strong generalization performance on unseen data. To ensure a fair comparison, all algorithms were trained using the same experimental framework, identical training and testing datasets, consistent feature sets, and the same data balancing strategies.

The evaluated machine learning algorithms included Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, Linear Support Vector Machine (Linear SVM), and Multi-Layer Perceptron (MLP). In addition, three deep learning architectures—Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN)—were implemented to investigate whether deep learning models could better capture the complex relationships among crime-related, spatial, temporal, socio-economic, weather, and mobility features.

To ensure consistency throughout the experiments, each algorithm was trained using the same training dataset and evaluated using the same independent testing dataset. Furthermore, algorithm-specific hyperparameters were optimized before the final evaluation so that each model was assessed under its most appropriate configuration. This standardized experimental setup ensured that performance differences reflected the capabilities of the algorithms rather than inconsistencies in the experimental design.

The comparison of algorithms was based on multiple classification performance metrics, including Accuracy, Precision, Recall, F1-score, Receiver Operating Characteristic Area Under the Curve (ROC-AUC), and Precision–Recall Area Under the Curve (PR-AUC). Since the arrest prediction problem involves an imbalanced binary classification dataset, greater emphasis was placed on Recall, F1-score, ROC-AUC, and PR-AUC. These metrics provide a more meaningful assessment of a model’s ability to correctly identify arrest cases while maintaining an appropriate balance between false positive and false negative predictions.

In addition to predictive performance, the algorithms were compared based on their robustness,

computational efficiency, and ability to generalize to unseen data. Traditional machine learning algorithms offered advantages in terms of faster training time and simpler model interpretation, whereas ensemble learning methods demonstrated improved predictive performance by combining multiple decision trees. Deep learning models were also evaluated to determine their effectiveness in learning complex patterns from the integrated multi-source dataset.

Based on the comprehensive comparative analysis, the final model was selected using a combination of quantitative performance metrics and overall model reliability rather than relying on a single evaluation criterion. The selected model demonstrated superior predictive capability while maintaining a balanced trade-off between precision and recall, making it the most suitable approach for arrest prediction within the framework.

The detailed experimental results, comparative performance analysis, and justification for selecting the final predictive model are presented in the subsequent chapter. These results include a comprehensive comparison of all evaluated algorithms using the selected classification metrics and supporting visualizations.